Topics Covered:

- Introduction to Exponential Representation
- Exponential Representation of Rotations

Additional Reading:

- MLS Chapter 2, Section 2.2; LP 3.2.3

Exponential Representation of Group Motion / Displacements

In linear control, we often describe the dynamics of a linear system as:

$$\dot{x} = Ax$$

where x is our state variable of size n and A is a matrix of size $n \times n$. In the case of a linear time-invariant system, the solution to this differential equation is:

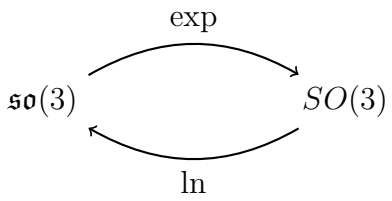
$$x(t) = e^{At}x(0)$$

where e^{At} is known as the matrix exponential. We define this matrix exponential as:

$$e^{At} = \sum_{n=0}^{\infty} \frac{(At)^n}{n!}$$

In this class, we are going to use this same Exponential Mapping to map between Lie Groups ($SO(3)$ and $SE(3)$) and their corresponding Lie Algebras ($\mathfrak{so}(3)$ and $\mathfrak{se}(3)$).

We will go over these expressions later, but in general you can think of the following map:

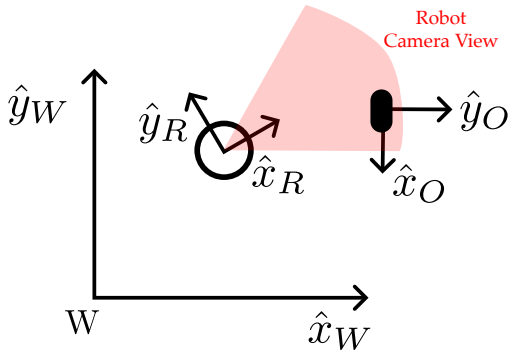


Here, exp is called the exponential map, and ln (the inverse operation) is called the Logarithm. These correspond to the matrix exponential and matrix logarithm operations respectively.

In general, exp and ln are useful for translating between finite transformations (rotations or rigid body motions) and their infinitesimal generators (like angular velocities and twists), allowing smooth and efficient computations in 3D space.

Example of Exponential Representation

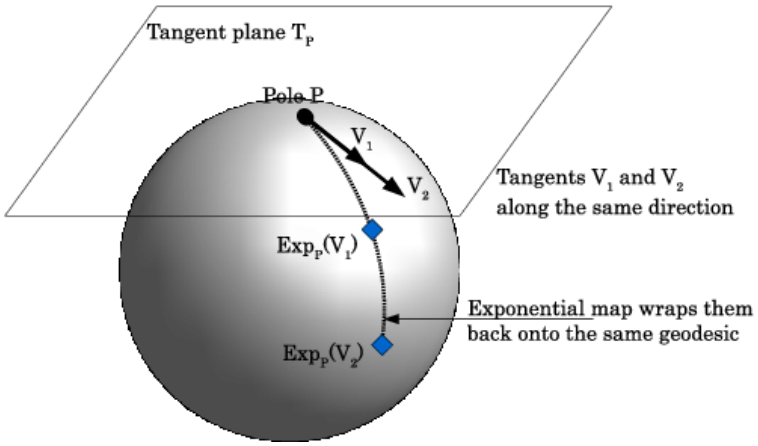
Consider the example with a mobile robot in the 3D world.



The mobile robot obtains measurements of its orientation using an inertial measurement unit (IMU) that provides us with angular velocities. We can then compute the exponential map at each time step to obtain how the orientation of the robot changes over time. For example, if the angular velocity readings of the robot are $\omega(t) = [0, 0, 0.1]^\top$:

$$R(0 + \delta t) = \exp([\omega]_\times \delta t) R(0)$$

A visual representation of this mapping is the following (we will discuss in class):



One main benefit of using these exponential representations is that they allow us to smoothly interpolate between different orientations and to integrate rotational dynamics over time.

Method	Pros	Cons	When to Use
Exponential Map	Accurate for integrating rotations, natural for Lie groups	Requires matrix exponentiation, more complex mathematically	Advanced robotics and control applications
Quaternions	Compact, no gimbal lock, efficient for integration	Slightly more abstract than Euler angles	Real-time systems (e.g., drones, IMU fusion)
Euler Angles	Intuitive, simple for small rotations	Subject to gimbal lock, less efficient for complex rotations	Basic applications, small-angle motions

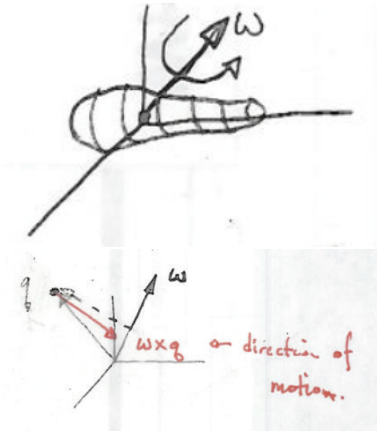
Table 1: Comparison of Rotation Representation Methods

We will cover exp, ln for each group and their properties. In this lecture, we will cover the special orthogonal group, with the following lecture covering the remaining two. Outline:

1. $SO(3)$ and $\mathfrak{so}(3)$
2. $SE(3)$ and $\mathfrak{se}(3)$
3. $SE(2)$ and $\mathfrak{se}(2)$

Exponential Representation for $SO(3)$ and $\mathfrak{so}(3)$

In the context of rotations, $SO(3)$ represents the special orthogonal group in 3D space, which consists of 3x3 orthogonal matrices with determinant 1 that represent 3D rotations. The Lie algebra $\mathfrak{so}(3)$ represents the set of 3x3 skew-symmetric matrices, which correspond to angular velocities (or "infinitesimal rotations").



consider a point on a body undergoing rotation:

$$\dot{q}(t) = \omega \times q(t) = [\omega]_\times q(t) \quad (1)$$

here, ω is the axis of rotation, while $[\omega]_\times$ is the axis represented as a matrix
If we integrate this velocity, we get:

$$q(t) = e^{[\omega]_\times t} q(0)$$

WAIT: what is $e^{[\omega]_\times t}$? It is the solution to the differential equation in (1).

From $\mathfrak{so}(3)$ to $SO(3)$ using the Exponential Map

The matrix exponential function $\exp : \mathfrak{so}(3) \to SO(3)$ maps a skew-symmetric matrix (which encodes a rotation axis and angle) to a rotation matrix. This is often combined with some "units of time", τ. This operation essentially "exponentiates" the infinitesimal rotation to get a finite rotation.

Mathematically:

$$R(\omega, \tau) = e^{[\omega]_\times \tau} \in SO(3)$$

Later we will derive a closed form for this computation via Rodrigues' rotation formula.

From $SO(3)$ to $\mathfrak{so}(3)$ using the Logarithm

The matrix logarithm function $\ln : SO(3) \to \mathfrak{so}(3)$ maps a rotation matrix back to a skew-symmetric matrix, which corresponds to the angular velocity or axis-angle representation of the rotation.

Mathematically:

$$[\omega]_\times = \ln(R) \in \mathfrak{so}(3)$$

Computing the Exponential Map for Rotations

The definition of the exponent of a matrix is:

$$e^{\mathrm{A}} = \sum_{n=0}^{\infty} \frac{1}{n!} \mathrm{A}^n \qquad \text{(from Taylor series for } e^{\mathrm{A}}\text{)}$$

$$e^{[\omega]_\times \tau} = \sum_{n=0}^{\infty} \frac{1}{n!} ([\omega]_\times \tau)^n$$

If rotated for τ units of time, then:

$$R(\omega, \tau) = e^{[\omega]_\times \tau}$$

Can we compute $e^{[\omega]_\times \tau}$ without requiring the infinite series expansion definition? Yes! The closed-form answer is *Rodrigues' formula*. Here is where we are headed:

Rodrigues' Formula

For a unit vector $\hat{\omega}$ ($\|\hat{\omega}\| = 1$), with $W = [\hat{\omega}]_\times$ and rotation angle θ:

$$e^{W\theta} = I + \sin(\theta)\, W + (1 - \cos(\theta))\, W^2$$

Notation. Throughout, $\hat{\omega}$ is a unit axis ($\|\hat{\omega}\| = 1$) and we write $W = [\hat{\omega}]_\times$. With a unit angular velocity, rotating for τ units of time sweeps out an angle of exactly τ radians, so we will write θ in place of τ.

A Concrete Example: Rotation About the z-Axis

Before deriving the formula in general, let's compute the series by hand for the simplest case, $\hat{\omega} = \hat{z} = [0, 0, 1]^\top$. We already know the answer should be $R_z(\theta)$.

$$W = [\hat{z}]_\times = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad W^2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad W^3 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -W,$$

$$W^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -W^2, \qquad W^5 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = W.$$

Notice that:

- The powers repeat (up to sign) after two steps:

  $$W^3 = -W, \quad W^4 = -W^2, \quad W^5 = W, \quad W^6 = W^2, \quad \dots$$

  Odd powers alternate between $\pm W$, and even powers alternate between $\pm W^2$.

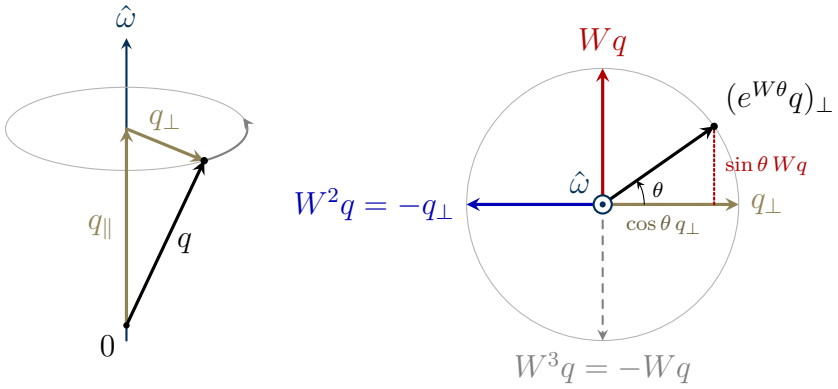
Plugging these into the series and adding up the matrices entry by entry:

$$\begin{aligned} e^{W\theta} &= I + \theta W + \frac{\theta^2}{2!}W^2 + \frac{\theta^3}{3!}W^3 + \frac{\theta^4}{4!}W^4 + \cdots \\ &= \begin{bmatrix} 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots & -\left(\theta - \frac{\theta^3}{3!} + \cdots\right) & 0 \\ \theta - \frac{\theta^3}{3!} + \cdots & 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = R_z(\theta). \end{aligned}$$

The exponential really does produce the rotation matrix we already knew. The only property of W that we needed was $W^3 = -W$. Does this hold for *every* unit axis $\hat{\omega}$, not just $\hat{z}$? To see why it does, let's look at what W and W^2 do geometrically.

What Do W and W^2 Do?

Take any point q and split it into a component along the axis, $q_\parallel = \hat{\omega}(\hat{\omega}^\top q)$, and a component perpendicular to it, $q_\perp = q - q_\parallel$. As the body rotates about $\hat{\omega}$, $q_\parallel$ stays fixed while $q_\perp$ sweeps out a circle.



side view looking down $\hat{\omega}$

- W **turns** $q_\perp$ **by** 90°**.** $Wq = \hat{\omega} \times q$ is tangent to the circle and has length $\|q_\perp\|$. Since $\hat{\omega} \times q_\parallel = 0$, W ignores the part of q along the axis.
- W^2 **turns it by another** 90°**.** Using the triple product identity $a \times (b \times c) = b(a^\top c) - c(a^\top b)$ and $\|\hat{\omega}\| = 1$:

  $$W^2 q = \hat{\omega} \times (\hat{\omega} \times q) = \hat{\omega}(\hat{\omega}^\top q) - q = q_\parallel - q = -q_\perp.$$

  So $W^2 = \hat{\omega}\hat{\omega}^\top - I$, which points from q straight back toward the axis. (For $\hat{\omega} = \hat{z}$, $\hat{\omega}\hat{\omega}^\top - I = \mathrm{diag}(-1, -1, 0)$, matching the example.)
- W^3 **is a** 180° **turn of** Wq**.** Applying W once more:

  $$W^3 q = W(-q_\perp) = -Wq_\perp = -Wq \quad \Longrightarrow \quad W^3 = -W.$$

This holds for any unit $\hat{\omega}$, so the pattern from the $\hat{z}$ example holds in general. Each application of W rotates the perpendicular part by 90°, which is exactly why sin and cos are going to appear.

Deriving Rodrigues' Formula

Using $W^3 = -W$, every power of W reduces to $\pm W$ or $\pm W^2$:

$$W^{2k+1} = (-1)^k W, \qquad W^{2k+2} = (-1)^k W^2, \qquad k = 0, 1, 2, \ldots$$

Substitute these into the series, then group the odd-power terms and the even-power terms:

$$\begin{aligned} e^{W\theta} &= I + \theta W + \frac{\theta^2}{2!}W^2 + \frac{\theta^3}{3!}W^3 + \frac{\theta^4}{4!}W^4 + \frac{\theta^5}{5!}W^5 + \frac{\theta^6}{6!}W^6 + \cdots \\ &= I + \theta W + \frac{\theta^2}{2!}W^2 - \frac{\theta^3}{3!}W - \frac{\theta^4}{4!}W^2 + \frac{\theta^5}{5!}W + \frac{\theta^6}{6!}W^2 + \cdots \\ &= I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) W + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots\right) W^2 \end{aligned}$$

Recall the series expansions for sin and cos:

$$\sin(\theta) = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots$$
$$\cos(\theta) = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots$$

The red series is exactly $\sin(\theta)$. The blue series is the cos series with its leading 1 removed and every sign flipped, i.e., $1 - \cos(\theta)$. Therefore:

$$e^{W\theta} = I + \underbrace{\left(\theta - \frac{\theta^3}{3!} + \cdots\right)}_{\sin(\theta)} W + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \cdots\right)}_{1-\cos(\theta)} W^2 = I + \sin(\theta)\, W + (1 - \cos(\theta))\, W^2$$

This expression is Rodrigues' Formula!

Interpreting each term (see the right panel of the figure). Applying the formula to a point q:

$$e^{W\theta} q = \underbrace{q}_{\text{start}} + \underbrace{\sin(\theta)\,(\hat{\omega} \times q)}_{\text{move along the tangent}} + \underbrace{(1 - \cos(\theta))(-q_\perp)}_{\text{pull back toward the axis}} = q_\parallel + \cos(\theta)\, q_\perp + \sin(\theta)\,(\hat{\omega} \times q).$$

The axial part $q_\parallel$ is unchanged, and the perpendicular part moves around the circle by angle θ, just as a rotation should.

Sanity check. Plugging $W = [\hat{z}]_\times$ into Rodrigues' formula:

$$I + \sin\theta \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + (1 - \cos\theta) \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = R_z(\theta). \checkmark$$

Aside: what if $\|\omega\| \neq 1$? Write the angular velocity as a unit axis times a speed, $\omega = \|\omega\|\,\hat{\omega}$. Rotating at speed $\|\omega\|$ for time τ sweeps out the angle $\theta = \|\omega\|\tau$, so

$$[\omega]_\times \tau = [\hat{\omega}]_\times\, \theta.$$

Now $\hat{\omega}$ is a unit vector, so we can apply Rodrigues' formula directly with $W = [\hat{\omega}]_\times = [\omega]_\times / \|\omega\|$:

$$e^{[\omega]_\times \tau} = I + \frac{[\omega]_\times}{\|\omega\|} \sin(\|\omega\|\tau) + \frac{[\omega]_\times^2}{\|\omega\|^2}(1 - \cos(\|\omega\|\tau))$$

Next, can we show that $e^{[\omega]_\times \tau} \in SO(3)$ (i.e., that $e^{[\omega]_\times \tau}$ produces a valid rotation matrix)?

Well, $R \in SO(3) \implies R^\top R = I$ and $\det(R) = 1$

a) $(e^{[\omega]_\times \tau})^\top (e^{[\omega]_\times \tau}) = e^{[\omega]_\times^\top \tau} e^{[\omega]_\times \tau} = e^{-[\omega]_\times \tau} e^{[\omega]_\times \tau} = I$

b) $\det(e^{[\omega]_\times \tau}) = e^{\mathrm{Tr}[\omega]_\times \tau} = e^0 = 1$

Thus, $e^{[\omega]_\times \tau} \in SO(3)$ is true. But what about the other way around?

The exponential map is surjective onto $SO(3)$ (Proposition 2.5 from MLS)

Proposition 1. *Given $R \in SO(3)$, there exists an $\hat{\omega} \in \mathbb{R}^3$ where $\|\hat{\omega}\| = 1$, and a $\theta \in \mathbb{R}$ such that $R = \exp([\hat{\omega}]_\times \theta)$.*

Note that this is like saying that there exists a function taking in an R and returning a pair $(\hat{\omega}, \theta)$ such that $R = \exp([\hat{\omega}]_\times \theta)$ where $\|\hat{\omega}\| = 1$.
This function will be called the logarithm and will be denoted by ln,

$$(\hat{\omega}, \theta) = \ln(R), \quad \text{where } \|\hat{\omega}\| = 1.$$

If we allow for $\|\omega\| = \theta$, then we will write

$$\omega = \ln R$$

The pair $(\hat{\omega}, \theta)$ are not necessarily unique!

Proof of Proposition

Proof. The proof is a little nasty and constructive.

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

Note: only 3 of these entries are uniquely determined, $(\hat{\omega}, \theta)$ has only 3 unique variables too.

$$e^{[\hat{\omega}]_\times \theta} = I + [\hat{\omega}]_\times \sin(\theta) + [\hat{\omega}]_\times^2 (1 - \cos(\theta))$$

$$\implies \text{let } s_\theta = \sin(\theta), c_\theta = \cos(\theta), v_\theta = (1 - \cos(\theta))$$

$$= \begin{bmatrix} 1 - v_\theta((\hat{\omega}^2)^2 + (\hat{\omega}^3)^2) & \hat{\omega}^1\hat{\omega}^2 v_\theta - \hat{\omega}^3 s_\theta & \hat{\omega}^1\hat{\omega}^3 v_\theta + \hat{\omega}^2 s_\theta \\ \hat{\omega}^1\hat{\omega}^2 v_\theta + \hat{\omega}^3 s_\theta & 1 - v_\theta((\hat{\omega}^1)^2 + (\hat{\omega}^3)^2) & \hat{\omega}^2\hat{\omega}^3 v_\theta - \hat{\omega}^1 s_\theta \\ \hat{\omega}^1\hat{\omega}^3 v_\theta - \hat{\omega}^2 s_\theta & \hat{\omega}^2\hat{\omega}^3 v_\theta + \hat{\omega}^1 s_\theta & 1 - v_\theta((\hat{\omega}^1)^2 + (\hat{\omega}^2)^2) \end{bmatrix}$$

$$= \begin{bmatrix} (\hat{\omega}^1)^2 v_\theta + c_\theta & \hat{\omega}^1\hat{\omega}^2 v_\theta - \hat{\omega}^3 s_\theta & \hat{\omega}^1\hat{\omega}^3 v_\theta + \hat{\omega}^2 s_\theta \\ \hat{\omega}^1\hat{\omega}^2 v_\theta + \hat{\omega}^3 s_\theta & (\hat{\omega}^2)^2 v_\theta + c_\theta & \hat{\omega}^2\hat{\omega}^3 v_\theta - \hat{\omega}^1 s_\theta \\ \hat{\omega}^1\hat{\omega}^3 v_\theta - \hat{\omega}^2 s_\theta & \hat{\omega}^2\hat{\omega}^3 v_\theta + \hat{\omega}^1 s_\theta & (\hat{\omega}^3)^2 v_\theta + c_\theta \end{bmatrix}$$

Equating matrix elements should give the answer. It will suffice to equate enough coefficients to find $\hat{\omega}$ and θ.

a) examine the trace.

$$\begin{aligned} \mathrm{Tr}(R) &= r_{11} + r_{22} + r_{33} \\ \mathrm{Tr}(e^{[\hat{\omega}]_\times \theta}) &= 1 + 2\cos(\theta) \\ \implies \quad 1 + 2\cos(\theta) &= \mathrm{Tr}(R) \\ \implies \quad \cos(\theta) &= \frac{\mathrm{Tr}(R) - 1}{2} \\ \implies \quad \theta &= \cos^{-1}\left(\frac{\mathrm{Tr}(R) - 1}{2}\right) \end{aligned}$$

This solution can be zero, and can go to $\pm 2\pi n$ and still give the same answer.
If $\theta = 0$, then $\hat{\omega}$ can be anything.
If $\theta \neq 0$, then need to find $\hat{\omega}$.
b) examine R.
Well, if we look at the off-diagonal terms of $R - R^\top$ and $e^{[\hat{\omega}]_\times \theta} - (e^{[\hat{\omega}]_\times \theta})^\top = e^{[\hat{\omega}]_\times \theta} - e^{-[\hat{\omega}]_\times \theta}$, we have

$$\begin{aligned} r_{32} - r_{23} &= 2\hat{\omega}^1 \sin(\theta) \\ r_{13} - r_{31} &= 2\hat{\omega}^2 \sin(\theta) \\ r_{21} - r_{12} &= 2\hat{\omega}^3 \sin(\theta) \end{aligned}$$

$\implies$

$$\hat{\omega} = \frac{1}{2\sin\theta} \begin{Bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{Bmatrix}$$

This formula fails at $\theta = \pi$: there $\sin(\theta) = 0$ and R is symmetric, so we get $0/0$. Instead, plugging $\theta = \pi$ into Rodrigues' formula gives $R = I + 2[\hat{\omega}]_\times^2 = 2\hat{\omega}\hat{\omega}^\top - I$, so $R + I = 2\hat{\omega}\hat{\omega}^\top$. Every nonzero column of $R + I$ is therefore parallel to $\hat{\omega}$, and normalizing one gives $\hat{\omega}$ up to sign. Both signs are valid, since rotating by π about $\hat{\omega}$ or $-\hat{\omega}$ gives the same R. ■

So, now that we got the answer, does it make sense?
Well, first off,

$$\mathrm{Tr}(R) = \sum \lambda_i$$

or the sum of eigenvalues, what are these like? $det(R) = \prod \lambda_i = 1$ or $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 1$. One eigenvalue is $\lambda_1 = 1$, which leaves $\lambda_2 \cdot \lambda_3 = 1$. The others are two complex conjugates $\lambda \cdot \lambda^\star = 1$

$$\begin{aligned} \mathrm{Tr}(R) &= 1 + \lambda + \lambda^\star \\ &= 1 + 2\mathrm{Re}(\lambda) \end{aligned}$$

$\Longrightarrow$

$$1 + 2\text{Re}(\lambda) = 1 + 2\cos(\theta)$$

Therefore, a solution is possible.

In summary:

Given $\hat{\omega} \in \mathbb{R}^3$ with $\|\hat{\omega}\| = 1$, or $\omega \in \mathbb{R}^3$ without $\|\omega\| = 1$:

$$\exp([\hat{\omega}]_\times \theta) = I + \sin(\theta)[\hat{\omega}]_\times + (1 - \cos(\theta))[\hat{\omega}]_\times^2 \qquad \text{(Unit Axis Formula)}$$

$$\exp([\omega]_\times \tau) = I + \frac{[\omega]_\times}{\|\omega\|}\sin(\|\omega\|\tau) + \frac{[\omega]_\times^2}{\|\omega\|^2}(1 - \cos(\|\omega\|\tau)) \qquad \text{(General Formula)}$$

These can be related through $\hat{\omega} = \omega/\|\omega\|$ and $\theta = \|\omega\|\tau$.

Given $R \in SO(3)$, there exists $(\exists)$ $\hat{\omega} \in \mathbb{R}^3$, $\theta \in \mathbb{R}$ $(\|\hat{\omega}\| = 1)$, defined by

$$\theta = \cos^{-1}\left(\frac{\text{Tr}(R) - 1}{2}\right) \begin{cases} \theta = 0, & \hat{\omega} \text{ is arbitrary} \\ 0 < \theta < \pi, & \hat{\omega} = \frac{1}{2\sin(\theta)} \begin{Bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{Bmatrix} \\ \theta = \pi, & \hat{\omega} = \pm\frac{c_k}{\|c_k\|}, \quad c_k = \text{any nonzero column of } R + I \end{cases}$$

and is written as

$$(\hat{\omega}, \theta) = \ln(R)$$

or if $\|\omega\|$ can be non-unit (with an implicit duration of $\tau = 1$)

$$\omega = \ln R$$

In this case $\|\omega\|$ encodes the amount of rotation, as θ did previously. Then:

$$\|\omega\| = \cos^{-1}\left(\frac{\text{Tr}(R) - 1}{2}\right)$$

$$\frac{\omega}{\|\omega\|} = \frac{1}{2\sin(\|\omega\|)} \begin{Bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{Bmatrix}$$